

List Data Type:

List is a collection of homogenous data item or heterogeneous data items enclosed between the pair of square brackets.

- **Homogenous data item:** it is a group of values consists of same type of data items.

E.X: numbers = [1, 2, 3, 4]

- **Heterogeneous data item:** it is a group of values consists of different type of data items.

E.X: data = [1, 2, 'hello', 3+7j, True]

Syntax to create a list:

Variable = [val1, val2, val3.....]

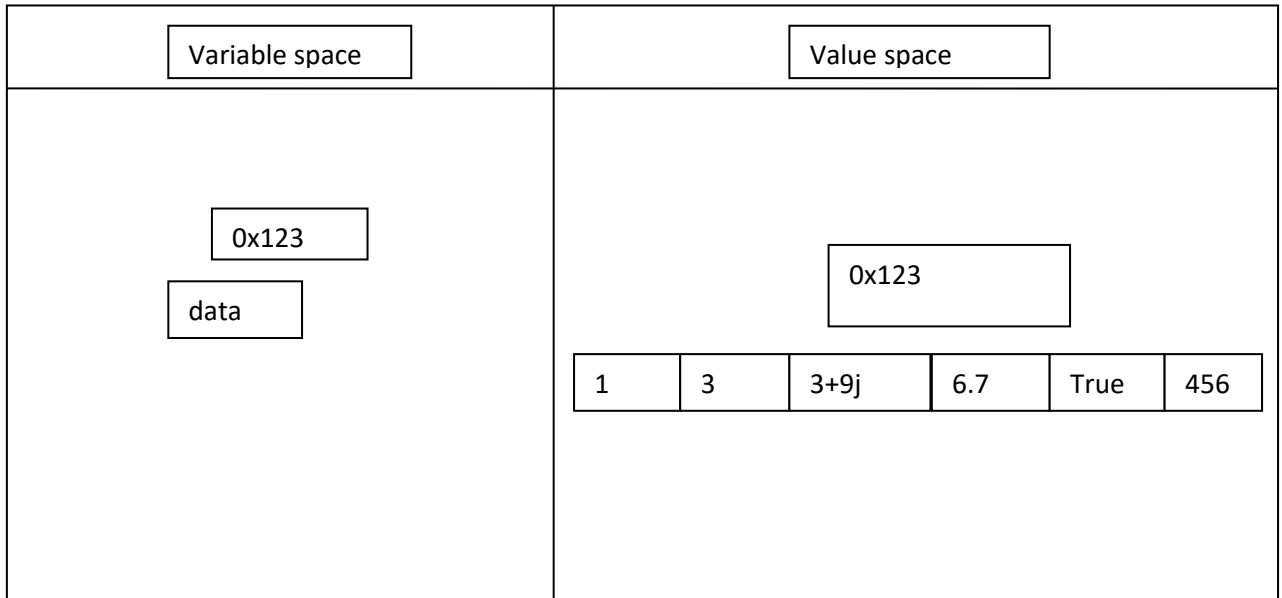
- The list values are separated by the ',' operator.
- The default value of the list is [] which is internally equal to bool False.

Memory allocation the list data type:

- As soon as , the control gets to know variable creation process it divide the memory into two space
 - Variable space
 - Value space
- When control gets to know that the value is collection it create the layer of memory in the value space.
- Layer will be divided into number of the blocks, which is exactly equal to the length of the collection.
- Address will be given to the created layer and that address will be stored with respect to the variable name in variable space.

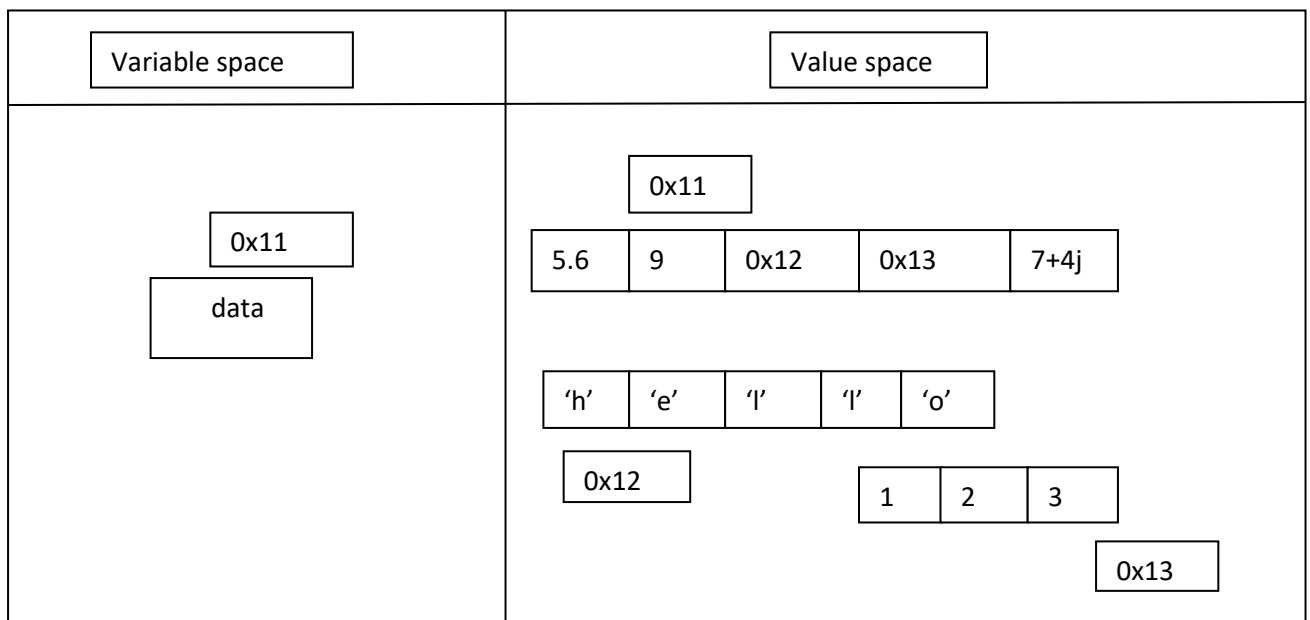
E.X:1

data = [1, 3, 3+9j, 6.7, True, 456]



E.X:2

data = [5.6, 9, 'hello', [1, 2, 3], 7+4j]



Attributes of list data type:

1. **Append ()**: It is a function, which is used to add a new item to the last position of an existing list collection.
 - `Variable.append(value)`
2. **Insert ()**: It is a function, which is used to add a new item to the specified position of an existing list collection.
 - `Variable.insert(position, value)`
3. **Count ()**: It is a function, which is used to count the number of occurrence of the specific item in the list.
 - `Variable.count(value)`
4. **Index ()**: It is a function, which is used to get position the specified item.
 - `Variable.index(value)`
5. **Remove ()**: It is a function, which used to removes the first occurrence of element or value specified.
 - `Variable.remove(value)`
6. **Pop ()**: It is a function, which used to removes the element of last position.
 - `Variable.pop()`
 - `Variable.pop (index)` → It is a function, which used to removes the element of specified position.
7. **Clear ()**: It is a function, which is used to clear all the elements present in the list.
 - `Variable.clear()`
8. **Sort ()**: It is a function, which is used to sort the list, in order the sort the list, it should be homogeneous. The sorting happens in original list itself.
 - `Variable.sort()`
9. **Extend ()**: It extend the existing list with given sequence.
 - `Variable.extend()`
10. **Copy ()**: It is a function, which is used to copy the values present in one variable to another variable.
 - `Variable.copy()`

Since it is possible to modify the value present inside the given list, we can call it as “mutable collection”.